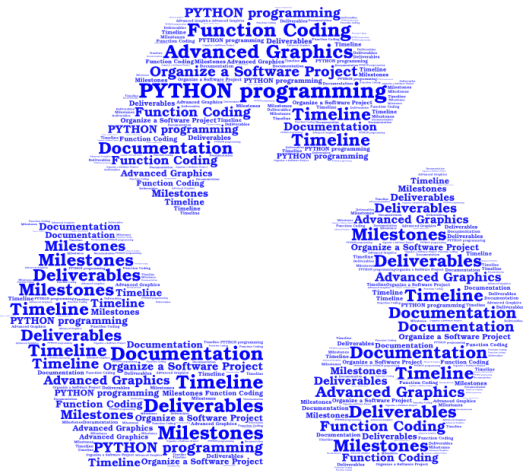




Python Programming: Work Package 2 – Solar Panel

Lecture: Modelling an Simulation

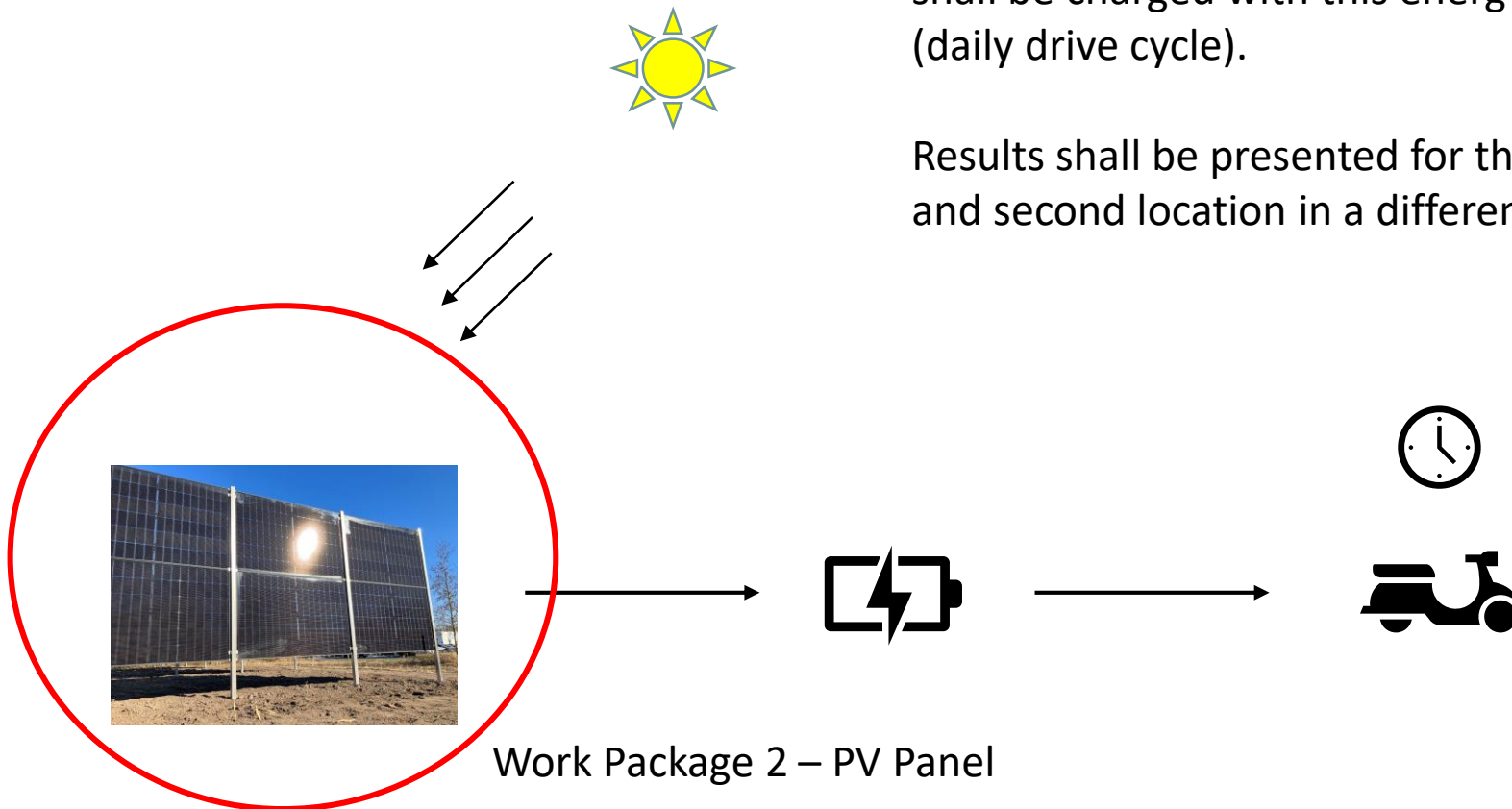
This is a compulsory elective module
of
**Technology of Circular
Economy**
(Master of Engineering)

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Senior Scientist

Overview of the project

Design a PV system which collects energy from the sun and stores it in a battery. An electric scooter shall be charged with this energy for commuting to work (daily drive cycle).

Results shall be presented for the city of Emden and second location in a different country.



Work Package 2 – PV Panel

Project Plan

		week						
Work package		1	2	3	4	5	6	
	Planning	*	M 1					
WP 1	Scooter							
WP 2	Solar panel							
WP 3	Battery				*	M 2		
WP 4	System Design							
WP 5	Calculate results						*	M 3
	Write report							
	List of Deliverables			List of Milestones				
	D 1	Software requirements		M 1	Plan ready			
	D 2	Scooter model		M 2	All models ready			
	D 3	PV model		M 3	Results ready			
	D 4	Battery model						
	D 5	System model						

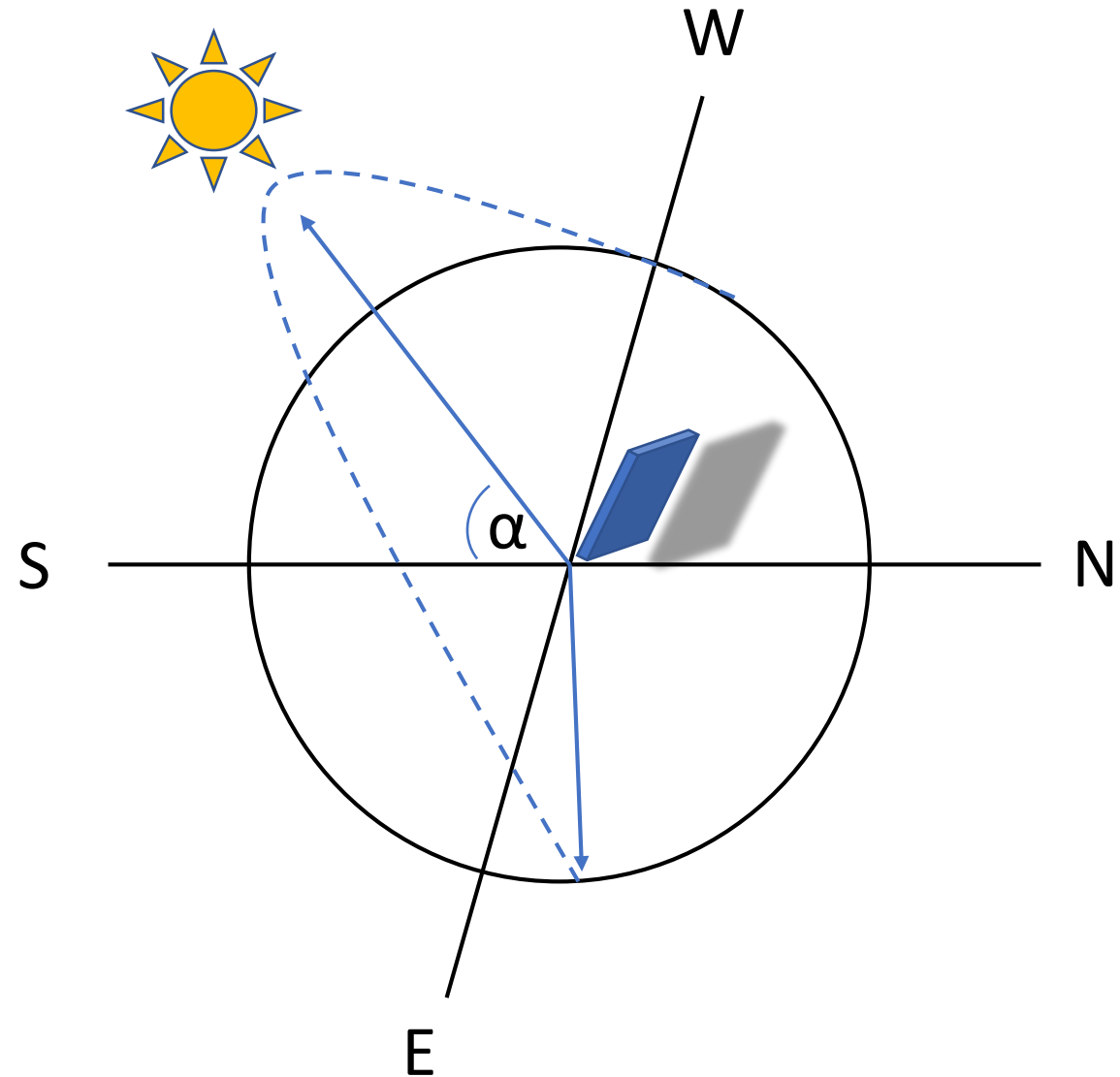
Energy from the sun

Photovoltaic panels collect radiation from the sun and convert it to electric energy.

The movement of the sun changes with the seasons of the year. In the left, you see a sketch for the situation in Emden during summer. Daytime is longer than night time, the sun is not exactly rising in the east and not exactly setting in the west.

The maximum and minimum daylight times for the city of Emden are 17 hours in summer and 7 hours in winter.

For this lecture we will not calculate the movement of sun. We rather use recorded data.



PVGIS website: https://re.jrc.ec.europa.eu/pvg_tools/en/

PVGIS = **P**hotovoltaic **G**eographical Information **S**ystem

The PVGIS website is provided by the European Commission and offers free and open access to meteorological data (radiation data) for photovoltaic system performance calculation. Presently, there are three key features.

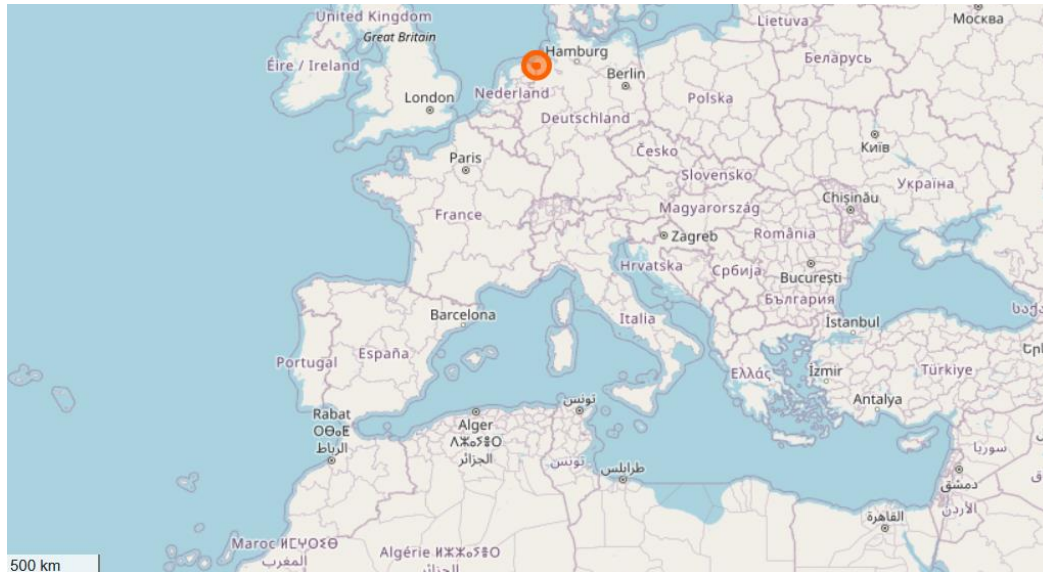
- Radiation data at different timely resolution (hourly, daily, monthly)
- Simulation tool that generates the power output based on: PV system power, location, surface tilt, azimuth.
- Radiation data and power production for a 'typical meteorological year' at the given location.

Emden/ Germany

Emden urban area has about 50 000 inhabitants.

Emden became an important port city in the 16th century under the rule of the Netherlands.

During World War II Emden was bombed heavily because of the production of submarines in the port area.

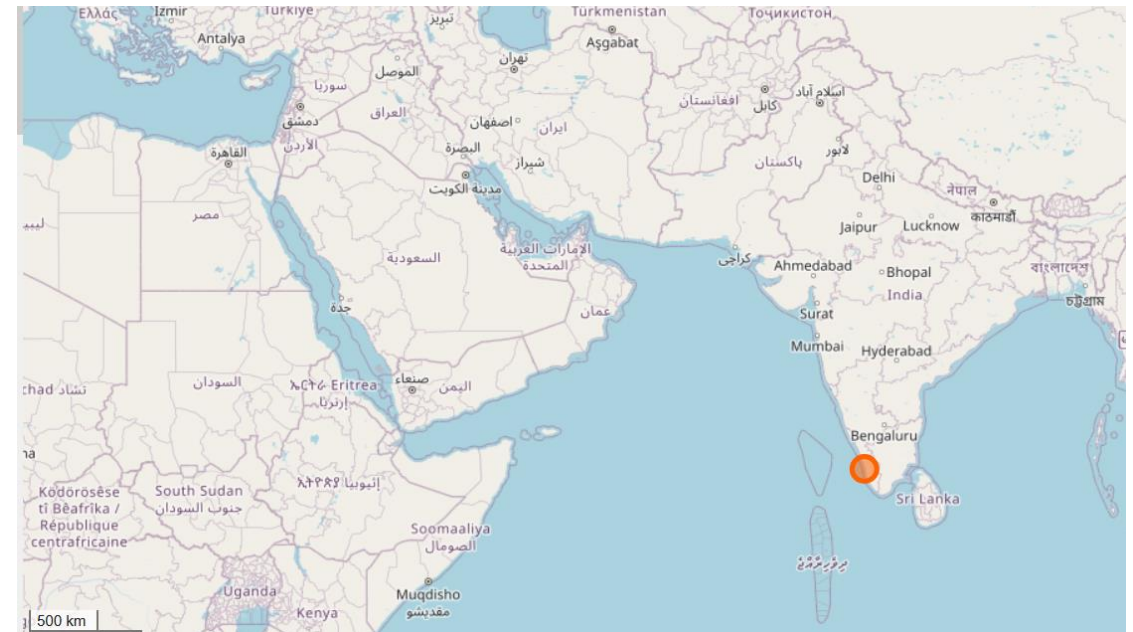


Two locations

Kochi/ India

Kochi urban area has 2.1 million inhabitants.

The port had great historical importance for trading spices. It connected Europe and the Arabic peninsula with China for a long period, reaching back to the European Greek and Roman empires.



Two locations



Work package 2 / WP 2: Solar

Input:

1. Location of PV: **S**
2. Size of the PV , Number of PV: **M**
3. Angle: **S**
4. Time duration per day: **C**
5. Maximum capacity: **M**
6. Sun data: **S**

Output:

1. generation of energy per day: **M**

Assumptions:

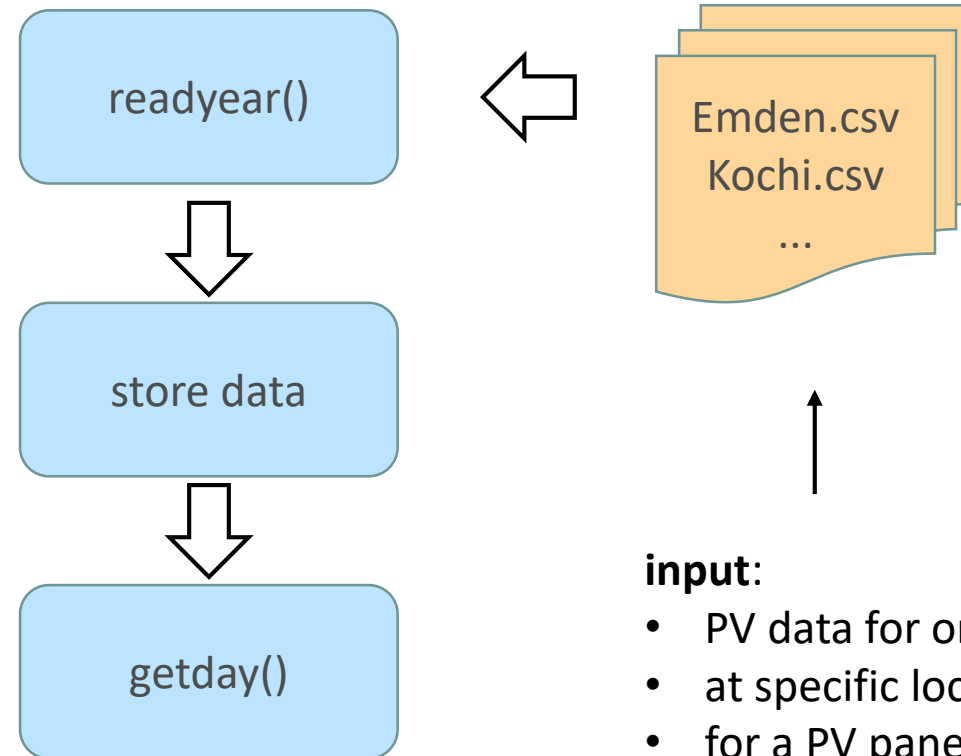
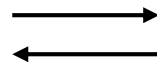
1. Sunlight per day: **C / W**
2. Energy losses: **W**

Deliverable D2: Generation of energy demand for the PV per day

Outlook: final structure

We have a set of PV data (2 files).
It would be good to read the files only
once, since they do not change.

input: day
output: produced energy



input:

- PV data for one year
- at specific location
- for a PV panel of 1 kW peak

The development of the model will be done in these steps.

- 1) Retrieve data from the PVGIS website
- 2) Reading a data file
- 3) Converting text to numbers
- 4) Collect numbers in lists
- 5) Calculate sums per day
- 6) Restructure code with function to be called from outside
- 7) Add a function to retrieve values for a specific day.

(This is what we finally need.)

Rooftop installation in Emden/ Germany

Latitude (decimal degrees): 53.368

Longitude (decimal degrees): 7.183

Orientation south (azimuth = 0 degree)

Slope (roof top) 23 degree

Nominal power 1.0 kW peak

System losses 14%

Rooftop installation in Kochi/ India

Latitude (decimal degrees): 10.026

Longitude (decimal degrees): 76.329

Orientation south (azimuth = 0 degree)

Slope (roof top) 23 degree

Nominal power 1.0 kW peak

System losses 14%

Modelling

PVGIS website: https://re.jrc.ec.europa.eu/pvg_tools/en/

We use the data for the year 2023. Choose hourly data and export as *.csv file.

(You can download the data files from the Moodle system of the lecture.)

Emden-2023.csv / Header information (top of file)

```
1 Latitude (decimal degrees): 53.368
2 Longitude (decimal degrees): 7.183
3 Elevation (m): 3
4 Radiation database: PVGIS-ERA5
5
6
7 Slope: 23 deg.
8 Azimuth: 0 deg.
9 Nominal power of the PV system (c-Si) (kWp): 1.0
10 System losses (%): 14.0
11 time,P,G(i),H_sun,T2m,WS10m,Int
12 20230101:0030,0.0,0.0,0.0,14.97,9.66,0.0
13 20230101:0130,0.0,0.0,0.0,14.48,10.28,0.0
```

Location

PV system

data

There are 11 lines, before the data starts.

Emden-2023.csv / Footer information (bottom of file)

```
8770 20231231:2230,0.0,0.0,0.0,8.06,6.76,0.0
8771 20231231:2330,0.0,0.0,0.0,7.7,6.76,0.0
8772
8773 P: PV system power (W)
8774 G(i): Global irradiance on the inclined plane (plane of the array) (W/m2)
8775 H_sun: Sun height (degree)
8776 T2m: 2-m air temperature (degree Celsius)
8777 WS10m: 10-m total wind speed (m/s)
8778 Int: 1 means solar radiation values are reconstructed
8779
8780
8781
8782 PVGIS (c) European Union, 2001-2025
```

Information about data columns



There are 11 lines after the data is completed.

Step 2: reading a file

Remember from a past lecture how to open a data file.

```
9  infile = 'Emden-2023.csv' # file name
10
11  myin = open(infile, 'r') # open a text file in read
12  mytext = myin.readlines() # here, we read the enti
13  myin.close()
14
15  # remove header line from list / there are 11 lines
16  del mytext[0:11]
17
18  # remove lines at the end / there are also 11 lines
19  b = len(mytext)
20  a = b - 11
21  del mytext[a:b]
```

Step 3: converting text to numbers

Remember the split() function from past lecture.

```
26 | pall = []           # create a list to collect powe
27 |
28 | for a in mytext:
29 |     b = a.split(',') # split at comma
30 |     p = float( b[1] ) # convert second co
31 |     pall.append(p)    # add to list
32 |
33 | # check list
34 | print( len(pall) )    # there should be 8760
35 |
```

The development of the model will be done in these steps.

- 1) Retrieve data from the PVGIS website 
- 2) Reading a data file 
- 3) Converting text to numbers 
- 4) Collect numbers in lists
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- 6) Restructure code with function to be called from outside
- 7) Add a function to retrieve values for a specific day.

(This is what we finally need.)

How to write the solar program?

Step 4: compute sum for each day

How can you do that?

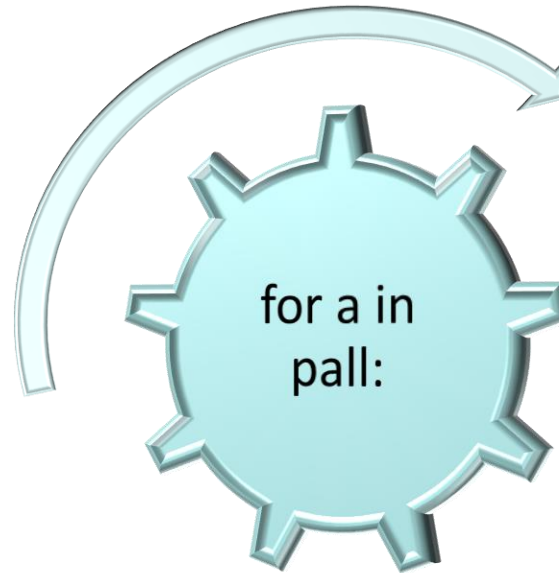
pall = list with power for all hours

How to write the solar program?

Step 4: compute sum for each day

How can you do that?

pall = list with power for all hours



- add to sum
- count to 24 h of a day

'if' test

- check, if 24 h are reached
- add sum to list

Step 4: compute sum for each day

How can you do that?

pall = list with power for all hours

```
35 pday = []
36
37 h = 0
38 p = 0
39 for a in pall:
40
41     p = p + a           # sum power
42     h = h + 1          # sum hour
43
44     if h == 24:        # this is one day
45         pday.append(p) # store sum for day
46         p = 0
47         h = 0
48
49 print( len(pday) )    # check the list
```

output should be 365

The development of the model will be done in these steps.

- 1) Retrieve data from the PVGIS website 
- 2) Reading a data file 
- 3) Converting text to numbers 
- 4) Collect numbers in lists 
- 5) Calculate sums per day 
- 6) Restructure code with function to be called from outside
- 7) Add a function to retrieve values for a specific day.
(This is what we finally need.)

Step 5: Restructure code with function to be called from outside

Here, the code that has been developed until now is shifted into a function **def readyear()**

You can download this step from the Moodle system as 'step05.py'.

```
12  infile = 'Emden-2023.csv'      # file r
13  pday = []                      # sum of
14
15  def readyear(infile):
16      # function to read file with dat
17      global pday
18
19      if len(pday) > 0:          # if the
20          del pday[0:]           # delete
21
22      myin = open(infile, 'r')    # c
23      mytext = myin.readlines()  # h
24      myin.close()
25
```

Step 6: Add a function -> getday()

Until now, the data is read from file, the sum for each day is calculated and stored into a list. Here, we add a function to yield the result for a specific day of the year.

```
54  
55 def getday(day):  
56     # function to retrieve sum of p  
57     global pday  
58  
59     day = day - 1 # internal count  
60     p = pday[day]  
61     return p
```

The development of the model will be done in these steps.

- 1) Retrieve data from the PVGIS website 
 - 2) Reading a data file 
 - 3) Converting text to numbers 
 - 4) Collect numbers in lists 
 - 5) Calculate sums per day 
 - 6) Restructure code with function to be called from outside 
 - 7) Add a function to retrieve values for a specific day. 
- (This is what we finally need.)

What now – we have achieved everything. Is something missing?

Step 7: Testing the model

In theory, we have finished the model development. Now, it would be good to test the output. Let us calculate the solar power generation per month in order to see the trend over one year.

How can we do that? Our model provides the power production for each day of a year (365). Now, we need to find out, which days belong to which month.

You can download the file step07.py from the Moodle system.

Now, we have the starting number for each month.
How can we get the sum for each month?

-> We will use the built in slice function for lists.

```
7  days = []
8  days.append(0) # This serves as starting point
9  days.append( 31 ) # January 31 days
10 days.append( 28 ) # Febr 28 days
11 days.append( 31 ) # Mar 31 days
12 days.append( 30 ) # Apr 30 days
13 days.append( 31 ) # May 31 days
14 days.append( 30 ) # Jun 30 days
15 days.append( 31 ) # Jul 31 days
16 days.append( 31 ) # Aug 31 days
17 days.append( 30 ) # Sep 30 days
18 days.append( 31 ) # Oct 31 days
19 days.append( 30 ) # Nov 30 days
20 days.append( 31 ) # Dec 30 days
21
22 mon = []
23 x = 0
24 for a in days:
25     x = x + a
26     mon.append(x)
```

The string 'hello'



Positions inside:

[0] [1] [2] [3] [4]



Built in functions:

Length: `b = 'hello'`
 `len(b)` → returns length of string, here = 5

Slice: extract text by position
 `b[0:2]` or `b[:2]` → returns text from position 0 up to 2 (but not including 2), here = 'he'
 `b[2:]` → returns text from position 2 to end, here = 'llo'

There are many more, but today we need only these two. Check out what else you can find for strings ...

Step 8: Calculate sum per month

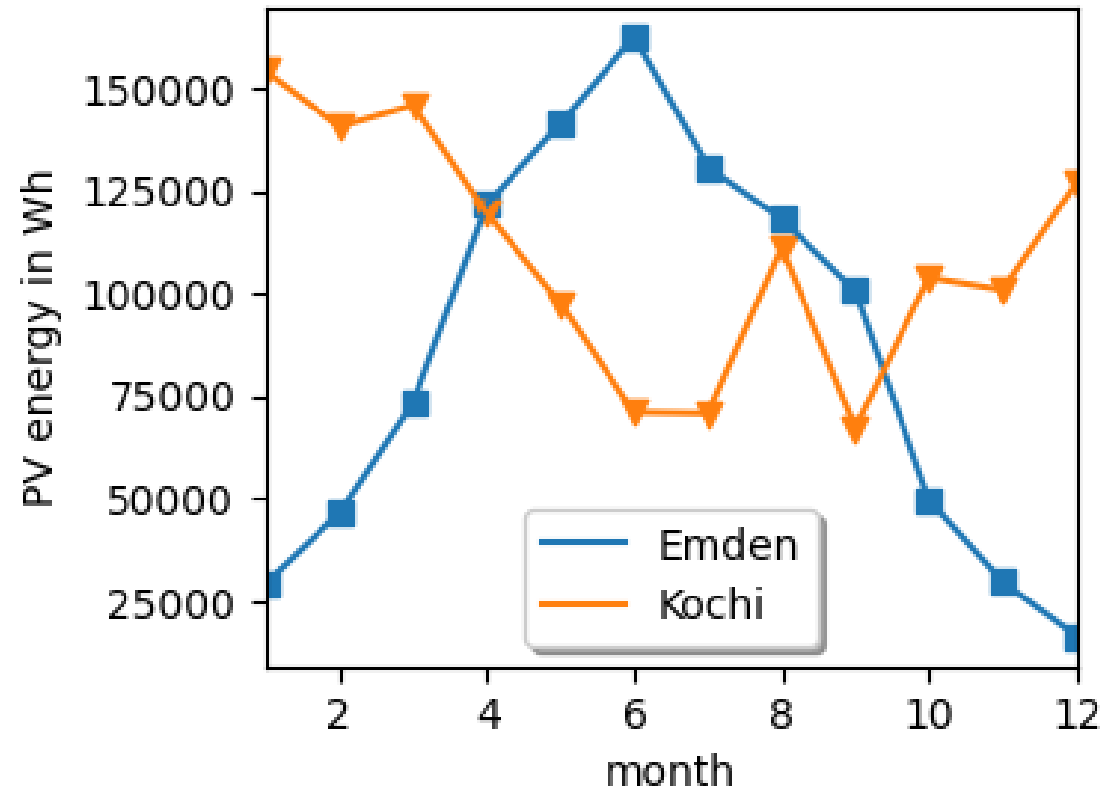
```
28 solar.readyear(file1) # initialize model, read data
29 year1 = []           # store results
30
31 x = 0
32 while x < 12:
33     a = mon[x]         # this marks the first day of the month
34     b = mon[x+1]       # this marks the first day of the NEXT month
35     x = x + 1
36
37     c = solar.pday[a:b] # cut out only the days for one month
38     p = 0
39     for d in c:         # sum over one month
40         p = p + d
41
42     year1.append(p)     # store sum in list
43
44 # check the result - should be 12 months
45 print( len(year1) )
```

Step 9: Print sum per month to check results

```
44 | # check the result per month
45 | x = 0
46 | for a in year1:
47 |     x = x + 1
48 |     print(x, a)
49 |
```

You can print it into a text file and create a plot with EXCEL.
We will learn in one of the next lectures, how to do this in Python.

With the testing program we can generate a profile of solar power per month for the two cities Emden and Kochi.



Results of the test – reality check

Sorry, I could not find sunshine data for Kochi (Cochin), but here is the data for another city in Kerala, Munnar.

<https://weather-and-climate.com/average-monthly-hours-Sunshine,munnar,India>

Daily hours of sunshine

